EduNest

Personalized Learning & Skill Development Platform

Short Project Note | Version 1.0 | October 2026

1. Project Overview

EduNest is a personalized e-learning platform designed to solve a major weakness of conventional learning platforms: they provide content, but often do not help students understand what they actually know, where they are weak, what they should learn next, or how to apply their knowledge.

> **Core idea: EduNest focuses on learning outcomes, not just course completion.**

2. Problem Statement

Students using online learning platforms commonly face information overload, lack of personalized guidance, difficulty identifying weak concepts, passive learning, poor revision habits, inconsistent study schedules, limited feedback, and difficulty connecting theoretical knowledge with practical skills.

Most platforms primarily show metrics such as “70% course completed”. However, completing a course does not necessarily mean that a student has mastered its concepts.

> **EduNest addresses this gap by continuously analyzing topic-level performance and guiding students toward what they need to learn, practice and revise.**

3. Product Vision

To create a learner-centered platform that converts educational content into measurable knowledge and practical skills through personalized learning, assessment, targeted revision and real-world challenges.

4. Target Users

Students: Learn subjects, identify weaknesses, practice, revise and build practical skills.

Instructors: Create courses, lessons, quizzes and practical challenges and monitor student performance.

Admin: Manage users, courses and basic platform moderation.

5. Main Differentiator

> **Traditional LMS: Course → Lesson → Quiz → Certificate**

> **EduNest: Goal → Diagnostic → Knowledge Profile → Personalized Path → Learn → Practice → Assess → Analyze → Revise → Apply**

6. Core Features

Feature	What it does
Diagnostic Assessment	Tests the student's existing knowledge before learning starts.
Knowledge Profile	Shows topic-level strengths and weaknesses.
Personalized Learning Path	Recommends which topics the student should learn or review next.
Lessons	Provides structured explanations, examples, key points and estimated study time.
Practice Questions	Lets students practice individual concepts before major assessments.
Quiz & Analysis	Evaluates performance and identifies weak topics instead of only showing total marks.
Topic Mastery	Tracks mastery for individual concepts rather than only overall course progress.
Revision System	Creates revision tasks for weak or previously learned concepts.
Daily Learning Plan	Suggests manageable activities for the student's current learning session.
Skill Map	Visually shows what the student actually knows across topics.
Practical Challenges	Converts theoretical knowledge into real-world tasks or mini-projects.
Notes & Bookmarks	Allows students to save personal notes and important lessons.
Instructor Dashboard	Shows course performance and difficult topics for instructors.

7. Adaptive Learning Logic

EduNest does not need machine learning for the academic MVP. A rule-based recommendation engine is sufficient and easier to explain and implement.

Example:

Mastery < 40% → Weak → Recommend lesson + practice + revision
Mastery 40–69% → Needs Practice → Recommend practice + revision
Mastery 70–89% → Strong → Allow progression + revision
Mastery ≥ 90% → Mastered → Recommend advanced challenge

Example mastery calculation: Diagnostic 20% + Practice 20% + Quiz 40% + Revision 20%.

8. Student Learning Flow

Register → Select Goal → Enroll in Course → Diagnostic Test → Knowledge Profile → Personalized Learning Path → Lesson → Practice → Quiz → Performance Analysis → Revision → Skill Map → Practical Challenge

9. Technology & Architecture

Frontend: HTML5, CSS3 and JavaScript

Backend: Spring Boot REST API

Database: MySQL

Development approach: REST-based communication between frontend and backend.

Important constraint: No React, Python, external AI API or other required technology for the MVP.

High-Level Architecture

Student / Instructor
↓
HTML + CSS + JavaScript
↓ REST API
Spring Boot
↓
Services / Business Logic / Security
↓
Spring Data / Repository
↓
MySQL

10. Main Database Entities

Entity	Purpose
users	Student, instructor and admin accounts.
courses / topics / lessons	Stores course structure and learning content.
enrollments	Tracks which students are enrolled in courses.
questions / quizzes / quiz_attempts	Stores assessments and student attempts.
topic_mastery	Stores topic-level learning mastery.
revision_schedule	Stores revision tasks and due dates.
learning_sessions	Tracks study sessions and duration.
challenges / submissions	Stores practical tasks and student submissions.
notes / bookmarks	Stores student-specific learning references.

11. Key Student Dashboard

The dashboard should answer four questions immediately:

What should I learn? → Personalized recommendation
What am I weak at? → Skill map
What should I revise? → Revision queue
What should I do today? → Daily learning plan

12. Example Student Experience

A student takes a Java diagnostic test and receives: Variables 90%, Loops 85%, Arrays 65%, OOP 42%, Inheritance 28%. EduNest identifies Inheritance as weak and recommends OOP → Classes → Inheritance before advanced topics. After a quiz, the platform updates mastery, schedules revision and eventually recommends a practical challenge such as building a Student Management System.

13. MVP Scope

The first working version should focus on the adaptive-learning core rather than trying to build a large commercial LMS.

Module	MVP Requirements
Authentication	Registration, login and role management.
Student	Course discovery, enrollment, diagnostic test, learning path, lessons, practice, quizzes, mastery, revision, skill map and practical challenge.
Instructor	Create courses, topics, lessons, quizzes and challenges; view student performance.
Backend	Spring Boot REST APIs, authentication, business logic and adaptive recommendation rules.
Database	MySQL persistence for users, courses, assessments, mastery and learning activity.

14. Development Roadmap

Phase 1: Project setup, MySQL database, Spring Boot structure and authentication.
Phase 2: Course, topic, lesson and enrollment system.
Phase 3: Diagnostic assessment, practice and quiz system.
Phase 4: Topic mastery, weak-topic detection and personalized learning path.
Phase 5: Revision system and daily learning plan.
Phase 6: Skill map and practical challenges.
Phase 7: Instructor analytics and UI refinement.

15. Success Metrics

EduNest can be evaluated using diagnostic-to-final score improvement, topic mastery improvement, completion of recommended learning paths, revision effectiveness, practical challenge completion and learning-session activity.

16. Project Demonstration Flow

For a college presentation, demonstrate one complete learning journey:

> **Login → Java Course → Diagnostic Test → Knowledge Profile → Personalized Path → Weak Topic → Lesson → Quiz → Learning Diagnosis → Revision Recommendation → Updated Skill Map → Practical Challenge**

17. Final Product Positioning

EduNest should not be presented as simply another online course platform. It is a Personalized Learning & Skill Development Platform that helps students identify knowledge gaps, follow adaptive learning paths, revise weak concepts and demonstrate knowledge through practical challenges.

One-line definition:

> **“EduNest transforms content-based e-learning into a personalized learning cycle that identifies knowledge gaps, recommends what students should learn next, schedules revision and connects learning with practical application.”**

18. Recommended Initial Course

Java Programming — Fundamentals, Variables, Operators, Conditions, Loops, Arrays, Methods, Classes & Objects, Encapsulation, Inheritance, Polymorphism, Exception Handling, Collections and a Mini Project.

> **Core principle: Measure what the student knows, identify what they need, and guide them toward the next useful learning action.**